Trisomy 18 and 13

What are trisomies?

The term trisomy is used to describe the presence of three chromosomes, rather than the usual pair of chromosomes. For example, if a baby is born with three #21 chromosomes, rather than the usual pair, the baby would be said to have "trisomy 21." Trisomy 21 is also known as Down syndrome. Other examples of trisomy include trisomy 18 and trisomy 13. Again, trisomy 18 or trisomy 13 simply means there are three copies of the #18 chromosome (or of the #13 chromosome) present in each cell of the body, rather than the usual pair.

What are trisomy 18 and trisomy 13?

Trisomy 18 and trisomy 13 are genetic disorders that include a combination of birth defects. This includes severe intellectual disability, as well as health problems involving nearly every organ system in the body. Unfortunately, most babies born with trisomy 18 or 13 die by age 1. It is important to note that some babies with trisomy 18 or 13 do survive the first year of life. Therefore, these disorders are not universally fatal. In the absence of any immediate life-threatening problems, accurate predictions of life expectancy are difficult to make. There are a few reports of babies with trisomy 18 or 13 surviving to their teens, however, this is unusual.

Trisomy 18 is also called "Edwards syndrome," named after the physician who first described the disorder. Trisomy 13 is called "Patau syndrome," in honor of the physician who first described it.

What causes trisomy 18 and trisomy 13?

Usually, each egg and sperm cell contains 23 chromosomes (half the normal number i body cells). The union of these cells creates 23 matched pairs, or 46 total chromosomes at the time of fertilization. In this manner, a person receives exactly half of their genetic material from each biological parent. Sometimes, an error occurs when an egg or sperm cell is forming, causing it to have an extra chromosome #18 or #13 inside. If this egg or sperm cell contributes that extra chromosome 18 to the embryo, then trisomy 18 results. If this egg or sperm cell contributes that extra chromosome 13 to the embryo, then trisomy 13 results. The extra chromosome 18 or 13 can come from either the mother or father. The features of trisomy 18 and trisomy 13 result from having this extra copy of chromosome 18 or 13 in each of the body's cells.

Occasionally, the extra chromosome 18 or 13 is attached to another chromosome in the egg or sperm; this is called a translocation. This is the only form of trisomy 18 or 13 that can be inherited from a parent. Sometimes, a parent can carry a "balanced" rearrangement, where chromosome 18 or 13 is attached to another chromosome. However, since the parent does not have any extra or missing chromosome material, they are said to have a "balanced translocation" and they are usually normal and healthy. Rarely, mosaic trisomy 18 or 13 may occur when the error in cell division occurs after fertilization. These affected persons have some cells with an extra chromosome 18 or 13 and others with the normal number.

What types of problems do children with trisomy 18 typically have?

Babies with trisomy 18 appear thin and frail. They fail to thrive and have problems feeding. Trisomy 18 causes a small head size, with the back of the head (occiput) prominent. Ears are usually low-set on the head. The mouth and jaw are unusually small, and there is a shortened sternum (breastbone).

At birth, these babies are small for their age, even when delivered full-term, and have a weak cry. Their response to sound is decreased and there is often a history of infrequent fetal activity during the pregnancy. Most babies with trisomy 18 have heart defects. They clench their fists in a characteristic manner and fully extending their fingers is difficult. Joint contractures—where the

arms and legs are in a bent position, rather than relaxed—are usually present. The feet may be referred to as "rocker bottom," due to their curved shape.

Babies with trisomy 18 may also have spina bifida, eye problems, cleft lip and palate, and hearing loss. It is also common to see feeding problems, slow growth, seizures, high blood pressure, kidney problems, and scoliosis (curvature of the spine). In males, the testes sometimes fail to descend into the scrotum.

Most babies with trisomy 18 have problems that affect all parts of the body in some way. The majority of children with trisomy 18 will have most, but not all, of the health problems mentioned here. Heart problems, feeding difficulties, and an increased susceptibility to infection are factors which, most often, contribute to the death of these children.

What types of problems do children with trisomy 13 typically have?

Babies with trisomy 13 often have a low birthweight, even when born full-term. They have a small head, with a sloping forehead. Usually, there are major structural problems with the brain that are diagnosed shortly after birth. Often, the front of the brain does not divide properly, resulting in a condition called holoprosencephaly. This can cause changes in the development of the baby's face, where the eyes are close set, or the nose or nostrils are underdeveloped. Cleft lip and cleft palate are common in babies with trisomy 13.

Eye problems are common and the ears are low-set and unusual in shape. Sometimes, babies with trisomy 13 can have scalp abnormalities (cutis aplasia) which resemble ulcers. They can also have birthmarks that are purplish-red in color; the color is due to tiny blood vessels close to the skin (hemangiomas).

Many babies with trisomy 13 have extra fingers and toes (polydactyly). The feet may have prominent heels. In many cases, there are other health problems present at birth. These include heart defects, kidney problems, and/or an omphalocele (a condition in which some of the abdominal organs protrude through an opening in the abdominal muscles in the area of the umbilical cord). In males, the testes sometimes fail to descend into the scrotum. Females may have an abnormally shaped uterus, called a bicornuate uterus.

How are trisomy 18 and trisomy 13 diagnosed?

Since trisomy 18 and trisomy 13 each have a unique group of characteristics, a physician may be able to determine whether a baby has trisomy 18 or 13 simply by physical examination. To confirm the physical findings, a small blood sample can be taken and the chromosomes can be analyzed to determine the presence of an extra #18 or #13 chromosome.

Chromosomal abnormalities such as trisomy 18 and 13 can be diagnosed before birth by analyzing cells in the amniotic fluid or from the placenta. Fetal ultrasound during pregnancy can also give information about the possibility of trisomy 18 or 13. Ultrasound is not 100 percent accurate, since some babies with trisomy 18 and 13 may look the same on ultrasound as those without the syndrome. A chromosome analysis, whether performed on a blood sample, cells from the amniotic fluid, or placenta, is very accurate.

What is the risk for parents of a child with trisomy 18 or trisomy 13 having another child with trisomy 18 or 13?

In general, for women under 35 years of age, the chance of having another baby with trisomy 18 or 13 depends on several factors. The risk to having a baby with trisomy 18 or 13 does increase slightly with each added year of maternal age.

After birth, the physician usually takes a blood sample from a baby suspected of having trisomy 18 or 13, to perform a chromosomal analysis (called a karyotype). This confirms the physical findings of trisomy 18 or 13 and determines the underlying chromosomal abnormality. This information is

important in determining the risk in future pregnancies. (Translocation and mosaic trisomy 18 and 13 have different recurrence risks for future pregnancies). The physician may refer parents to a genetic physician or genetic counselor who can explain the results of chromosomal tests in detail. This includes what the recurrence risks may be in another pregnancy and what tests are available to diagnose chromosome problems before a baby is born.

Can trisomy 18 or trisomy 13 be cured or prevented?

There is no cure for trisomy 18 or trisomy 13. We are not certain how to prevent the chromosomal error that causes trisomy 18 and trisomy 13. To date, there is no scientific evidence that a parent could have done anything to cause or prevent the birth of their baby with trisomy 18 or 13.

What else should I know about trisomy 18 and trisomy 13?

Because many babies born with trisomy 18 and trisomy 13 may not live beyond the first few days or weeks of life, it is possible that parents may have to face the fact that the baby may never go home from the hospital.

Many times, parents are frightened and overwhelmed by all of the information associated with trisomy 18 and 13. Decisions surrounding the care of an infant with trisomy 18 and 13 are difficult and personal. There are many resources available to parents to help them during this time, including early intervention services, hospice care, social workers, the hospital chaplain or clergyman, and genetic counselors. Families who have or have had a baby with trisomy 18 or trisomy 13 are particularly helpful and supportive since they have experienced many of the same questions and emotions.

Down syndrome is a genetic condition. It is not an illness or a disease.

Our bodies are made up of millions of cells. In each cell there are 46 chromosomes. The DNA in our chromosomes determines how we develop. Down syndrome is caused when there is an extra chromosome. People with Down syndrome have 47 chromosomes in their cells instead of 46. They have an extra chromosome 21, which is why Down syndrome is also sometimes known as trisomy 21.

Although we know how Down syndrome occurs, we do not yet know why it happens. Down syndrome occurs at conception, across all ethnic and social groups and to parents of all ages. It is nobody's fault. There is no cure and it does not go away.

Down syndrome is the most common chromosome disorder that we know of. One of every 700-900 babies born worldwide will have Down syndrome, although this number is lower in Australia - see Down syndrome population statistics. Down syndrome is not a new condition. People with Down syndrome have been recorded throughout history.

People with Down syndrome have:

- some characteristic physical features
- some health and development challenges
- some level of intellectual disability.

Because no two people are alike, each of these things will vary from one person to another.

A test for Down syndrome can be carried out before a baby is born. Down syndrome is usually recognised at birth and is confirmed by a blood test. It was named after Dr John Langdon Down who first described it.

What does it mean to have Down syndrome? Back to top

Most of the young people growing up with Down syndrome today will lead quite ordinary lives in the community. Some people with Down syndrome may not need much help to lead an ordinary life, while others may require a lot of support.

Having an intellectual disabilityBack to top

Down syndrome is the most common cause of intellectual disability that we know of. Everyone who has Down syndrome will have some level of intellectual disability. There will be some delay in development and some level of learning difficulty. Because everyone is unique, the level of delay will be different for each person.

When a baby is born, there is no way to tell what level of intellectual disability the child may have. Nor can we predict the way in which this may affect a person's life. However, we do know that having Down syndrome will not be the most important influence on how that person develops and lives their life. Instead, what happens after birth will be much more important and family, environmental, cultural and social factors will shape their life, just like everyone else.

For many people with Down syndrome, speaking clearly can be difficult. Although a lot of people with Down syndrome speak fluently and clearly, many will need speech and language therapy to achieve this. Very often, people with Down syndrome can understand a lot more than they can express with words. This often means that their abilities are underestimated, which can make them feel frustrated.

Some people with Down syndrome will find it very difficult to develop language skills and speak clearly. This may be made worse by hearing loss.

Living an ordinary lifeBack to top

People with Down syndrome are not fundamentally different from anyone else.

They have the same needs and aspirations in life that we all do, including:

- a good place to live
- meaningful employment
- the opportunity to enjoy the company of friends and family
- intimacy
- having a role in our community.

However, achieving these goals is harder for people with Down syndrome than it is for everyone else. Many people with Down syndrome are likely to need some level of support to help them achieve the kind of life that most people take for granted.

In the past, many people with Down syndrome have not had the opportunity to develop to their full potential. Often, they have been separated from the rest of the community, living in segregated settings such as care institutions. Low expectations were placed on them and there were limited opportunities for learning and personal growth.

Today we recognise that growing up in families and communities, with the same rights and responsibilities as everyone else, is vital to the development of people with Down syndrome.

To be a part of a community you have to be in it. This means that people who have only experienced life in a segregated setting may find it difficult to be included in the general community. Life for people who grow up being included in families and communities will be very different than it has been for those who have always lived in care facilities.

Encouraging children with Down syndrome to go to a school with their peers from their community has many benefits. It opens the way for a smooth transition to adulthood and encourages meaningful inclusion in the community.

People with Down syndrome need opportunities to reach their full potential, like we all do. When given these opportunities, they become valued and productive members of their families and the community.

Being an individualBack to top

One of the greatest challenges that people with Down syndrome face is the attitudes of other people who do not understand what it means to have Down syndrome. Despite much change, many people still don't see the individual person. Instead they just see 'Down syndrome' and expect everyone with Down syndrome to be more or less the same.

People with Down syndrome are very different from each other, just as we are all different. Every person with Down syndrome is unique, with their own talents, abilities, thoughts and interests. And, like everyone else, people with Down syndrome have strengths and weaknesses. While one person may read very well but find basic mathematics difficult, another might be a first-class cook and live independently in the community, but will have to work hard to speak clearly. Family

passions, culture, interests and skills are also likely to be shared by people with Down syndrome, as they may be by other members of the family.

People with Down syndrome do not all look alike. In fact, people with Down syndrome look more like other people in their own family than they look like others with Down syndrome. Although there are some physical features associated with Down syndrome, there is large variation in how many of these features an individual may have. For some people, one feature may be very prominent while in another it may not exist at all. Importantly, the physical characteristics of Down syndrome that a person may have do not tell us anything about that person's intellectual ability.

Another common misconception is that all people with Down syndrome are happy and affectionate. People with Down syndrome experience all the same emotions as everyone else. They get happy, sad, embarrassed, frustrated, thoughtful and fall in and out of love, just as we all do. They may, however, find it difficult to express their feelings in words. This can lead to frustration and the expression of feelings through behaviours.

The Power of LanguageBack to top

Language can empower individuals and their families or reinforces limiting stereotypes. Words, and the way we communicate, shape how a person is accepted in the community and impacts on their quality of life. Terminology referring to disability has changed over the years. The commonly accepted terminology for the diagnosis is:

A person with Down syndrome; or a person who has Down syndrome.

Please avoid using out-dated and degrading terminology – today we refer to a person with an intellectual disability.

By using appropriate terminology, and encouraging others to do the same, you are actively seeing the person first and as more important as the disability.

It is also worth being mindful that:

- People do not ‘suffer from’ Down syndrome. Avoid other terms that imply this. For example, ‘a victim of’ or ‘afflicted by’.
- People with Down syndrome are not by definition ‘special’, ‘poor’ or ‘unfortunate’. They should be referred to with the same respect we expect people to show everyone.
- Statements which reinforce stereotypes, and suggest that everyone with Down syndrome is the same, may cause offence.

What Is Klinefelter Syndrome?

Klinefelter syndrome is a fairly common genetic condition found in males only.
Many boys with Klinefelter syndrome — also known as **XXY syndrome** — have no signs or symptoms, and some don't even know they have it until later in life.

The XXY condition that causes Klinefelter syndrome can't be changed, but medical treatment and working with therapists can help a boy's development and help lessen the condition's effects. Even without treatment, most boys with Klinefelter syndrome grow up to live productive, healthy lives.

What Causes Klinefelter Syndrome?

Boys who have Klinefelter syndrome are born with it. It's also called XXY because they have an extra X chromosome in most or all of their cells.

Usually, a person has 46 chromosomes in each cell, divided into 23 pairs, which includes two sex chromosomes. Half of the chromosomes are inherited from the father and the other half from the mother. The chromosomes contain genes, which determine an individual's characteristics, such as eye color and height. Boys typically have one X chromosome and one Y chromosome, or XY, but boys with XXY syndrome have an extra X chromosome, or XXY.

Klinefelter syndrome is not caused by anything the parents did or did not do. The disorder is a random error in cell division that happens when a parent's reproductive cells are being formed.

If one of these defective cells contributes to a successful pregnancy, the baby will have the XXY condition in some or all of his cells. Some boys will even have more than two X chromosomes (XXXY or XXXXY, for example), which increases the risk of severe symptoms and other health concerns.

When a boy is born with the XXY condition in only some of his cells, it's called **mosaic Klinefelter syndrome**. Often, boys and men with this condition can have milder signs and symptoms than others with the XXY condition.

What Are the Signs & Symptoms of Klinefelter Syndrome?

Not all boys with Klinefelter syndrome will have noticeable symptoms. Other boys can have symptoms that are physically apparent or problems with speech, learning, and development.

Babies with Klinefelter syndrome typically have weak muscles, reduced strength, and quiet personalities. They also can take longer to do things like sit up, crawl, walk, and speak.

Compared with other kids their age, boys with Klinefelter syndrome might have some or all of these symptoms:

- a taller, less muscular body
- broader hips and longer legs and arms
- larger breasts (a condition called gynecomastia)
- weaker bones
- a lower energy level

- smaller penis and testicles
- delayed or incomplete puberty (some boys won't go through puberty at all)
- less facial and body hair following puberty

What Problems Can Happen?

Klinefelter syndrome typically causes a boy's testicles to grow at a slower rate than those of other boys. It also prevents the testicles from producing normal amounts of sperm and the hormone **testosterone**.

Testosterone affects the way a boy develops both physically and sexually. Low hormone levels and problems with sperm production make it difficult or sometimes impossible for a boy with Klinefelter syndrome to father a child later in life.

Many boys with Klinefelter syndrome show symptoms related to their development of social and language skills. They may have trouble paying attention. A lot of boys learn to talk late or have trouble using words to express their emotions. They also can have trouble with things like learning to spell, read, and write.

Socially, they tend to have quiet personalities. They rarely cause trouble and are often more helpful and thoughtful than other boys. They're often shy and sensitive, and many are less self-confident and less active than other boys their age.

Most boys aren't likely to have major health problems, but the condition can bring some other challenges later in life. Klinefelter syndrome puts males at greater risk of breast cancer, some other cancers, and some other diseases like type 2 diabetes, varicose veins and problems with blood vessels, problems with sexual function, and osteoporosis (weak bones) later in life.

Most boys with Klinefelter syndrome can have sex when they become men, usually with the help of testosterone treatment. However, problems with their testicles prevent them from making enough normal sperm to father children.

The vast majority of men with Klinefelter syndrome are infertile and can't father a child the usual way. Options for becoming natural parents are limited, but fertility researchers are working on new treatments. By the time a child with Klinefelter syndrome is ready to become a dad, there may be new options available related to the extraction of sperm from the testicles.

How Is Klinefelter Syndrome Diagnosed?

Since Klinefelter syndrome can be hard to notice, many parents don't know their son has it until he grows up or shows delays in puberty. Sometimes, parents who are worried about their son's development consult a doctor, and the diagnosis reveals Klinefelter syndrome. This can help, because the earlier a boy is diagnosed with Klinefelter syndrome, the more effective the treatments usually are.

To diagnose cases of Klinefelter syndrome, doctors usually begin by asking about any learning or behavior issues and examining the boy's testicles and body proportions.

There are two main types of tests used in the diagnosis of Klinefelter syndrome:

1. Hormone testing, which is usually done by taking a blood sample to check for abnormal hormone levels.

2. A chromosome analysis, or karyotype, which is usually done on a blood sample. This test checks the number of chromosomes to see if the XXY syndrome is present.

How Is Klinefelter Syndrome Treated?

There's no way to change the XXY condition if a boy is born with it, but treatments can help relieve some of the symptoms. As with many conditions, beginning treatment early can greatly increase its effectiveness.

Testosterone replacement therapy (TRT) works by increasing a boy's testosterone levels into the normal range. Additional testosterone can help a boy with Klinefelter syndrome develop bigger muscles and a deeper voice, as well as promote growth of the penis and facial and body hair. It can also help improve bone density and reduce the growth of a boy's breasts. Testosterone therapy cannot increase the size of a boy's testicles or prevent or reverse infertility, however.
Educational support services can help boys and teens with Klinefelter syndrome keep pace in school. Many benefit from extra assistance when it comes to schoolwork. If your son has Klinefelter syndrome, let his teachers and school nurse know about his condition and see what kind of support is available. He may be eligible for an individualized education plan (IEP) or 504 education plan, which both can provide accommodations for kids with special needs.

Speech therapy and physical therapy can help boys with Klinefelter syndrome learn to speak, read, and write better, or improve muscle strength and coordination. Other forms of therapy include behavioral, mental health, and occupational therapy. These can help improve low self-confidence, shyness, and delayed social development.

Looking Ahead

Because boys with Klinefelter syndrome can have problems with schoolwork and sports, they may feel like they don't fit in with other kids their age. They're more likely to have low self-esteem, which can make things harder socially and academically. But by the time they're men, most will have normal social relationships with friends, family members, and others.

If your son struggles in school or has trouble making friends, talk to your doctor, school principal, or school counselor. Counselors and therapists can give boys practical skills to help them feel more confident in social settings. And many schools provide educational services or accommodations that can help your son succeed.

Turner Syndrome

What are monosomies?

The term monosomy is used to describe the absence of one member of a pair of chromosomes. Therefore, there is a total of 45 chromosomes in each cell of the body, rather than 46. For example, if a baby is born with only one X sex chromosome, rather than the usual pair (either two X's or one X and one Y sex chromosome), the baby would be said to have "monosomy X." Monosomy X is also known as Turner syndrome.

What is Turner syndrome?

Turner syndrome is a genetic disorder that occurs in girls. It causes them to be shorter than others and to not undergo normal puberty as they grow into adulthood. The severity of these problems varies among affected individuals. Other health problems may also be present involving the heart or renal system (for example, kidneys). Many of the health problems affecting girls with Turner syndrome can be managed or corrected with appropriate medical treatment. Turner syndrome occurs in approximately 1 in 2,000 to 2,500 females born.

The name "Turner syndrome" comes from the doctor, Dr. Henry Turner, who first described the collection of findings in 1938. It was not until 1959 that the cause of Turner syndrome (having only a single X chromosome) was identified.

What causes Turner syndrome?

Normally in reproduction, the egg cell of the mother and the sperm cell of the father start out with the usual number of 46 chromosomes. The egg and sperm cells undergo a cell division where the 46 chromosomes are divided in half and the egg and the sperm cells end up with 23 chromosomes each. When a sperm with 23 chromosomes fertilizes an egg with 23 chromosomes, the baby ends up with a complete matched set of 46 chromosomes, half from the biological father and half from the biological mother.

Sometimes, an error occurs when an egg or sperm cell is forming, causing it to have a missing sex chromosome. When this cell fails to contribute a sex chromosome to the embryo, so that there is only one X sex chromosome in the fetus, then Turner syndrome results. Having a single copy of a particular chromosome, rather than the usual pair, is called "monosomy." Turner syndrome is also known as "monosomy X." The missing sex chromosome error can occur in either the mother's egg cell or the father's sperm cell; however, it is usually an error that occurred when the father's sperm cell was forming. There is nothing known that a father could have done or not done that would have caused or prevented the sperm from having a missing sex chromosome. (The chance for Turner syndrome to occur is, therefore, not associated with the increasing age of the mother). The features of Turner syndrome result from having a missing X chromosome in each of the body's cells.

About half of all instances of Turner syndrome result from full monosomy X. Others have a mosaic pattern (two or more chromosome patterns in the cells). A small percentage of instances of Turner syndrome result from having the normal number of chromosomes (46 total), but missing a portion of the X chromosome. When only part of an X chromosome is missing (called a deletion), not all of it, girls with Turner syndrome usually have milder features of the syndrome. The features of Turner syndrome that are present depend on which part of the X chromosome they are missing.

What types of problems do girls with Turner syndrome typically have?

About half of girls born with Turner syndrome will have puffy hands and feet at birth, in addition to wideness and webbing of the neck. During a pregnancy, the doctor may have seen a structure called a "cystic hygroma" during fetal ultrasound. Cystic hygromas are fluid-filled sacs at the base

of the neck and they often go away prior to birth; but, in some cases, are still present in the newborn period.

Girls with Turner syndrome usually have a low hairline at the back of the neck, minor differences in the shape and position of the ears, a broad chest with widely spaced nipples, an increased number of small brown moles (nevi) on the skin, and deep-set nails. Short stature is the most visible feature of Turner syndrome. The average adult height of a woman with Turner syndrome is 4 feet, 8 inches. Most women with Turner syndrome are born with poorly formed or absent ovaries. Ovaries produce estrogen and without it, incomplete sexual development occurs. The usual signs of puberty—breast development, menstruation, and growth of pubic and axillary hair—do not occur without hormone therapy. The resulting infertility is not correctable, in most cases. Heart problems, kidney problems, and thyroid problems are also common in Turner syndrome and should be evaluated early on. About one in ten girls with Turner syndrome is born with coarctation of the aorta (constriction of the main artery leaving the heart), which sometimes needs to be corrected surgically. Other features which have been seen in Turner syndrome include feeding problems during infancy, middle ear infections, skeletal problems, and "cubitus valgus." Cubitus valgus basically describes a situation where a person standing with their arms at their sides, will have their elbows slightly bent. They are unable to keep their arms perfectly straight at their sides. Other reported medical findings in Turner syndrome include diabetes, dry skin, high blood pressure, thyroid problems, a small jaw, and a narrow high-arched palate (the top of the inside of the mouth).

Girls with Turner syndrome generally have normal intelligence. They tend to score higher on their verbal IQ than their nonverbal IQ, and may have problems in the areas of spatial perception and an increased incidence of specific learning disabilities.

How is Turner syndrome diagnosed?

When a girl is born with features suggestive of Turner syndrome, a small blood sample is usually taken and the chromosomes are analyzed to determine the absence of a sex chromosome. Sometimes, girls with Turner syndrome do not exhibit any problems as infants or children, and it is only when they fail to go through puberty that a doctor begins to suspect that Turner syndrome may be present. Again, not all girls with Turner syndrome exhibit all features described here. There is great variability. Some girls with Turner syndrome are diagnosed during the newborn period, while others are diagnosed during childhood and in their late teens.

Chromosomal abnormalities, such as Turner syndrome, can often be diagnosed before birth by analyzing cells in the amniotic fluid or from the placenta. Fetal ultrasound during pregnancy can also give information about the possibility of Turner syndrome, but ultrasound is not 100 percent accurate, and many babies with Turner syndrome may look the same on ultrasound as those without Turner syndrome. A chromosome analysis, whether performed on a blood sample, cells from the amniotic fluid, or placenta, is very accurate.

What is the risk of parents of a daughter with Turner syndrome having another daughter with Turner syndrome?

In general, in each subsequent pregnancy, the chance of having another baby with Turner syndrome would not be increased over the general population risk that affects every woman.

After birth, the doctor usually takes a blood sample from a baby suspected of having Turner syndrome to perform a chromosomal analysis (called a karyotype). This confirms the physical findings of Turner syndrome and determines the underlying chromosomal abnormality.
Your doctor may explain the results of the test to you or refer you to a genetic doctor or genetic counselor who can explain the results of chromosomal tests, as well as what tests are available to diagnose chromosome problems before a baby is born.

Can Turner syndrome be cured or prevented?

There is no cure for Turner syndrome; however, many of the more serious problems can be treated. For example, growth hormone and androgen therapy can be given to increase final adult height; hormone replacement therapy can be given so that girls will develop secondary sexual characteristics; coarctation (narrowing to an abnormal width) of the aorta can be surgically corrected if needed; and medications are available to treat high blood pressure, diabetes, and thyroid problems. Women with Turner syndrome can give birth to children, if desired, through the use of egg donation.